

DDM: Dense Dynamics World Action Modeling for Dynamics-Sensitive Robotic Manipulation

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Abstract—Video generative models serve robotics as action-conditioned world models and goal-conditioned world-action models. Although their interfaces differ, both usually assume one fixed dynamics regime, i.e., an invariant action-outcome mapping. When dynamics change without altering appearance, identical actions yield different futures and identical goals require different actions. Conditioning on past interactions is therefore a natural response. Raw histories record realized transitions, but do not explicitly separate nominal transition structure from the discrepancy relevant to adaptation. World-action models make this model-relative discrepancy observable by generating an explicit future alongside the action; once the action is executed, prediction and realization define a model-relative offset. The same residual construction provides a reference-centered interaction representation in WM mode. We therefore introduce DDM, a dynamics-aware video-diffusion framework that converts this model-relative prediction error into context. DDM-cond compares realized futures with a frozen dynamics-blind reference and aggregates persistent latent offsets into a residual dynamics descriptor, without dynamics labels or a predefined parameterization. DDM-pred complements this context by packing transient observations into the existing latent budget. A shared architecture family supports separately trained WM and WAM modes. We evaluate both modes in simulation on pushing with latent center-of-mass shifts and tossing with latent actuation variations. Across both domains, interaction conditioning improves dynamics-sensitive prediction and adapts generated actions. On pushing at $k = 8$, residual conditioning improves dynamics identification from 31.96% to 44.25% over matched history, while DDM raises open-loop action success by 12.38 percentage points over matched Cosmos Policy. By coupling both mechanisms, DDM makes latent dynamics actionable for prediction and action generation. Project page: ddm-robo.github.io.

I. INTRODUCTION

Video generative models increasingly support robotic manipulation in two complementary roles: action-conditioned world models (WM) predict how a scene will evolve, while goal-conditioned world-action models (WAM) jointly generate an action and the future it should realize [1]–[4]. Although their interfaces differ, both typically learn a single, fixed mapping between actions and outcomes. Dynamic manipulation breaks this assumption: latent changes in inertial or actuation dynamics can leave the scene visually unchanged while making the same action produce different futures and the same goal require different actions. Figure 1 exposes this ambiguity by replaying the same push from a bit-identical state under different dynamics. Across our counterfactual evaluation windows, final positions differ by a median of 7.97 cm and by as much as 19.61 cm. A dynamics-blind model must therefore reconcile incompatible transitions from identical visual evidence.

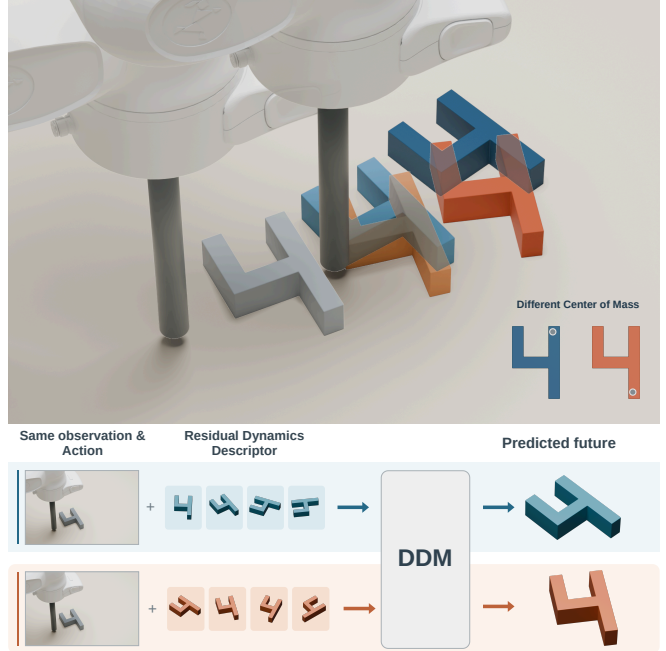


Fig. 1. **High-level idea of DDM for dynamics-aware prediction.** Given a current observation and action, a dynamics-aware world model should predict the future induced by the active, visually latent dynamics. *Top*: identical observation-action pairs produce different futures under distinct center-of-mass regimes. *Bottom*: DDM aggregates model-relative residuals from prior interactions into a residual dynamics descriptor (RDD), which conditions the same world model to predict the corresponding dynamics-consistent future. Colors denote regimes; thumbnails are schematic, not object-pose encodings. By representing what a dynamics-blind reference fails to predict, DDM turns prior interactions into actionable dynamics context without dynamics labels.

A common response is to condition policies or predictive models on past interactions [5]–[10]. Raw histories record realized transitions, but do not explicitly separate reference-predictable transition structure from the discrepancy relevant to adaptation. The conditioning network must therefore learn to discount appearance, action geometry, and nominal motion before extracting mismatch that persists across interactions.

World-action models make this model-relative discrepancy directly observable. Unlike conventional action-only policies, these video-generative policies emit an explicit future prediction alongside the generated action, externalizing the action-outcome relation encoded by the model. Once an action is executed, the offset from the predicted future to its realization exposes how that relation failed under the active regime. Aggregating the persistent structure of this model-relative error across interactions can therefore condition subsequent

action and future generation. The same construction applies in WM mode, where the recorded action is shared by the realized and reference futures. Video generation creates a second opportunity: the future target determines which dynamics evidence reaches the model. A terminal state may erase revealing motion after an object settles, whereas a dense trajectory preserves it.

We introduce DDM, a dynamics-aware video-diffusion framework built around these two observations. DDM-cond encodes latent offsets between realized futures and those predicted by a frozen dynamics-blind reference, then aggregates them into a residual dynamics descriptor (RDD) that modulates generation without dynamics labels. DDM-pred complements this model-relative context by packing observations across the prediction horizon into the existing future latent slot. We instantiate both components in separately trained WM and WAM instances of the same architecture, an interface we call *world action modeling*. Our contributions are:

- We pioneer video-generative dynamics-aware world action modeling, unifying dynamics-aware future prediction and action generation within a common generative framework.
- We introduce DDM-cond, a model-relative conditioning mechanism that distills persistent dynamics mismatch from prior interactions, and pair it with DDM-pred, a dense future objective that preserves transient dynamics evidence within the existing latent budget.
- We validate DDM on challenging matched-counterfactual push-digit and toss-cube benchmarks. Across latent inertial and actuation shifts, interaction conditioning improves dynamics-sensitive prediction, few-shot adaptation, and action generation over matched dynamics-blind models; push-digit further isolates the advantage of model-relative residuals over matched history.

II. RELATED WORK

Video models for robotic prediction and action generation. Action-conditioned video models, including the Cosmos backbone used here [11], predict future observations and enable planning through imagined rollouts [3], [4]. World-action models extend this interface by jointly generating actions and their anticipated outcomes [2]. This progression turns future prediction into an explicit account of the action-outcome relation encoded by the model. DDM uses that account as a reference for dynamics adaptation, while retaining separately trained WM and WAM modes within one architecture [1].

Dynamics conditioning and interaction-based adaptation. Dynamics-aware models have conditioned prediction or control on simulator parameters, spatial property maps, or learned dynamics latents [12]–[14]. Classical pushing mechanics relates mass distribution to the action-outcome map [15], [16]; learned systems infer latent dynamics from interaction or counterfactual prediction [5], [6], [17]–[20]. IIDA aggregates raw transitions for implicit identification [7],

whereas IPA Policy conditions one-shot action prediction on a partial probing interaction [9]. DyWA adapts a state-based WAM from history, while supervising its embedding with privileged geometry and dynamics features [8]. These methods establish interaction history as a powerful adaptation signal, but represent it either through privileged supervision or as absolute transition evidence. Direct history can also induce copycat or causal-confusion effects [21], [22]. Our IIDA+ and IPA Policy+ baselines scale the corresponding history-conditioning principles to pretrained WM and WAM backbones with k complete interactions. DDM receives the same observations, actions, and realized futures, but expresses each transition relative to a frozen reference model’s expected future. It therefore isolates the mismatch relevant to adaptation without estimating an explicit dynamics parameter.

Prediction residuals as adaptation signals. Prediction error is a classical identification signal [23] and has more recently supported ballistic correction, world-model adaptation, and residual dynamics learning [24]–[27]. These approaches use residuals primarily as correction signals for nominal predictions or learned transitions. DDM instead treats prediction error as a conditioning representation: it aggregates latent discrepancies from completed interactions before generating subsequent futures and actions. Residual structure that persists across interactions from one regime captures how its dynamics depart from the frozen reference, while aggregation attenuates transition-specific variation.

III. METHOD: DDM

A. Problem formulation

Let o_t be the visual observation at query time t , and let a_t denote the action representation executed from that state. In our controlled domains, a_t compactly parameterizes a manipulation primitive rather than a temporal action sequence. Let H denote the future-prediction horizon, and let $o_{t,H}$ be the corresponding future observation target assembled from frames in $(t, t + H]$. The frozen video tokenizer $\mathcal{E}$ maps the current observation and future target to $z_t = \mathcal{E}(o_t)$ and $z_{t,H} = \mathcal{E}(o_{t,H})$, respectively. When t and H are clear, we write $z^+ \equiv z_{t,H}$. In WAM mode, the visual goal marker is rendered directly into o_t before encoding; z_t therefore denotes a goal-marked current observation, rather than a separate observation-goal pair. Let ξ index the unobserved dynamics regime and let p_ξ denote its controlled transition distribution. The ambiguity motivating DDM is that different regimes induce different transitions even when the observation and action are identical:

$$z^+ \sim p_\xi(z^+ | z_t, a_t). \quad (1)$$

The robot neither observes ξ nor receives its identity as a label. Instead, the query is accompanied by an ordered history of k completed interactions from the same persistent regime. The i -th interaction contains its starting latent z_i , recorded action a_i , and realized future latent z_i^+ . DDM summarizes this history as a mode-specific residual dynamics descriptor (RDD) ψ_k^j , where $j \in \{\text{WM}, \text{WAM}\}$. Rather

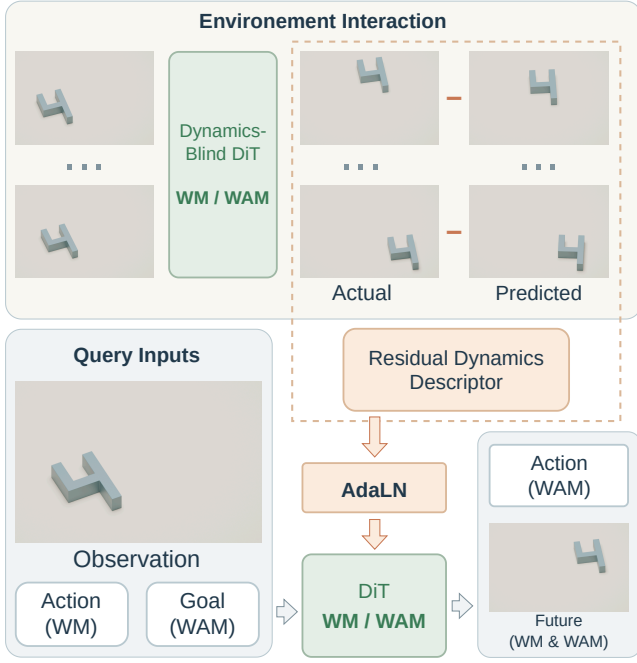


Fig. 2. **Residual dynamics conditioning in DDM.** For each prior interaction, a frozen dynamics-blind reference predicts a future and, in WAM, an action. Its latent future error is encoded with the starting latent and executed action, then aggregated across interactions into the RDD. For a new query, the descriptor modulates the DiT through AdaLN: WM predicts the future conditioned on the query action, whereas WAM jointly generates the action and future from the goal-marked observation. Thumbnails are schematics rather than decoded outputs. DDM thereby turns persistent reference error into actionable context for dynamics-responsive prediction and action generation.

than estimating ξ , the descriptor need only retain persistent model-relative mismatch useful for adapting prediction or action generation. We omit j when the mode is unambiguous; Sec. III-C defines the encoder and aggregation explicitly.

B. Architecture and modeling modes

Figure 2 gives an overview of the residual-conditioning interface shared by both modeling modes. Each mode instantiates a 2B-parameter Cosmos-Predict2 video diffusion transformer. Following the latent token injection design of Cosmos Policy [2], the normalized action representation is tiled over one video-latent frame. Its spatiotemporal VAE compresses four raw frames into one latent frame. The resulting sequence contains four components: a fixed-zero blank placeholder, current observation, action, and future target, with mode-specific masks; the placeholder is excluded from the loss.

Denoising follows the EDM formulation [28]. In Eq. (2), $\mathbf{x}_0$ is the clean sequence; m and ℓ are the conditioning and loss masks; $\mathbf{c}$ is the language embedding; and $\mathbf{n}$ is isotropic Gaussian noise at level σ :

$$\tilde{\mathbf{x}} = m \odot \mathbf{x}_0 + (1 - m) \odot (\mathbf{x}_0 + \mathbf{n}),$$

$$\mathcal{L}_{\text{DDM}} = \mathbb{E} \left[w(\sigma) \|\ell \odot (D_\theta(\tilde{\mathbf{x}}; \sigma, \mathbf{c}, \psi_k) - \mathbf{x}_0)\|_2^2 \right]. \quad (2)$$

where $w(\sigma)$ is the EDM weight and D_θ is the pre-conditioned denoiser. The masks separately specify observed

and supervised components; σ enters through adaptive layer normalization, $\mathbf{c}$ through cross-attention, and ψ_k through modulation. The expectation covers data and context draws, diffusion noise, and descriptor dropout.

The two modes capture how dynamics affect both the outcome of a given action and the action required for a desired outcome. Their factorizations are

$$\text{world model: } p_{\theta_{\text{WM}}}(z^+ | z_t, a_t, \psi_k), \quad (3)$$

$$\text{world-action model: } p_{\theta_{\text{WAM}}}(a_t, z^+ | z_t, \psi_k), \quad (4)$$

Mode-specific masks implement the two factorizations: WM conditions on the current observation and action and supervises the future, whereas WAM conditions on the goal-marked current observation and supervises the jointly generated action and future. We use *world action modeling* for this broader generative interface, rather than for a single jointly shared model: θ_{WM} and θ_{WAM} are optimized separately and use distinct frozen references and residual stores.

C. DDM-cond: Residual dynamics conditioning

A residual is meaningful only relative to a stable expectation. Each mode therefore uses a frozen dynamics-blind reference, Cosmos-Predict2 for WM and Cosmos Policy for WAM, from which we warm-start the corresponding DDM model. Training then jointly optimizes the encoder, modulation layers, and backbone. Both references are trained without interaction context and therefore marginalize over the training dynamics.

Let $f_{\bar{\phi}_{\text{WM}}}$ and $f_{\bar{\phi}_{\text{WAM}}}$ denote the distinct frozen references, and let $\mathcal{S}_{\text{WM}}$ and $\mathcal{S}_{\text{WAM}}$ be their fixed sampling operators. For interaction i ,

$$\hat{z}_i^{\text{ref}, \text{WM}} = \mathcal{S}_{\text{WM}}(f_{\bar{\phi}_{\text{WM}}}; z_i, a_i, \epsilon_0), \quad (5)$$

$$(\hat{a}_i^{\text{ref}, \text{WAM}}, \hat{z}_i^{\text{ref}, \text{WAM}}) = \mathcal{S}_{\text{WAM}}(f_{\bar{\phi}_{\text{WAM}}}; z_i, \epsilon_0), \quad (6)$$

$$\delta_i^j = z_i^+ - \hat{z}_i^{\text{ref}, j}. \quad (7)$$

Here $j \in \{\text{WM}, \text{WAM}\}$ and ϵ_0 is the fixed sampler-noise realization for the stored interaction; the checkpoint, sampler, and noise seed are shared across matched arms, defining a stable model-relative coordinate.

For a fixed WM reference, residualization is a deterministic, reference-centered reparameterization of the realized future; its role is to expose prediction mismatch explicitly before context aggregation, rather than to introduce additional observations.

For WM, the observed and predicted futures share a_i , so δ_i^{WM} is a matched-action transition discrepancy. WAM introduces a second source through its blind joint reference,

which jointly samples action and future from $p_{\bar{\phi}_{\text{WAM}}}(a, z^+ | z) = p(a | z)p(z^+ | z, a)$. Let the realized draw be $z_i^+ = F_\xi(z_i, a_i)$, and let $\bar{F}(z, a)$ denote a fixed-sampler conditional future coordinate under the second factor. Because the observed future follows a_i but the reference future accompanies $\hat{a}_i^{\text{ref}, \text{WAM}}$, defining $\nu_i = \bar{F}(z_i, \hat{a}_i^{\text{ref}, \text{WAM}}) - \hat{z}_i^{\text{ref}, \text{WAM}}$ yields the accounting

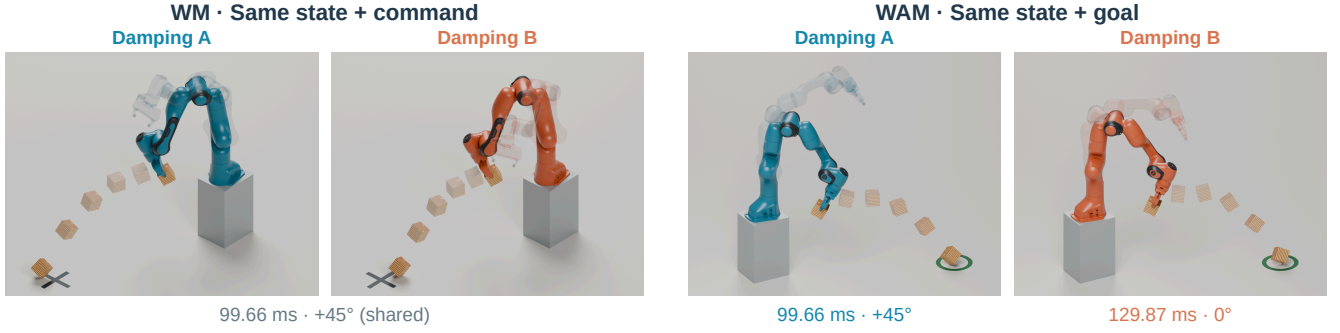


Fig. 3. **Cube tossing under latent actuation dynamics.** *Left pair (WM):* the same initial state and command produce different motion under two damping settings. *Right pair (WAM):* the same state and goal require different release-time/yaw commands. Crosses provide a fixed reference; rings mark the goal. Solid arms show release and faded arms show follow-through; colors identify damping. Grasp, release, and cube flight are schematized from recorded motion, ending at first contact. The paired construction exposes dynamics ambiguity in both forward prediction and goal-conditioned action generation.

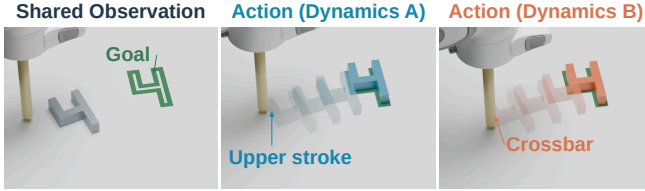


Fig. 4. **Goal-conditioned digit pushing.** Different hidden mass distributions require different push primitives to reach one target from the same observation. Faded and solid digits show motion and final pose; green outlines mark the goal. The shared observation and goal therefore leave the correct action ambiguous without dynamics context.

identity

$$\begin{aligned}\delta_i^{\text{WAM}} &= \Delta_i^{\text{trans}} + \Delta_i^{\text{act}} + \nu_i, \\ \Delta_i^{\text{trans}} &= F_\xi(z_i, a_i) - \bar{F}(z_i, a_i), \\ \Delta_i^{\text{act}} &= \bar{F}(z_i, a_i) - \bar{F}(z_i, \hat{a}_i^{\text{ref}, \text{WAM}}).\end{aligned}\quad (8)$$

Here Δ_i^{trans} is matched-action transition mismatch, Δ_i^{act} reflects the reference’s different action choice, and ν_i retains joint-sampling variation. DDM does not identify them separately: an individual δ_i^{WAM} is model-relative interaction evidence, not a pure estimate of ξ . Its aggregate ψ_k^{WAM} becomes dynamics context by retaining mismatch that persists across shared-regime interactions.

A mode-specific encoder E_j and projection P_j aggregate these residuals:

$$h_i^j = E_j(z_i, a_i, \delta_i^j), \quad \psi_k^j = P_j([b_1 h_1^j; \dots; b_k h_k^j]), \quad (9)$$

where h_i^j is the per-interaction embedding and $b_i \in \{0, 1\}$ masks absent entries. Each interaction field is projected to 256 dimensions before a 512-wide residual trunk; the k ordered, masked slots are concatenated and projected to the 256-dimensional descriptor. Selected interactions retain each task sampler’s deterministic order, shared across matched arms; any padding occupies trailing slots and is zero-masked. The descriptor is learned end-to-end through the denoising objective in Eq. (2); no auxiliary dynamics-label loss is used.

The matched history control replaces δ_i^j with z_i^+ in Eq. (9). We report it as IIDA+ in WM and as IPA Policy+ in WAM.

The resulting ψ_k modulates generation through two pathways in each transformer block rather than through the latent sequence. Let s index latent frames, $e_{\sigma, s}$ denote the noise-level embedding, and r_s the adaptive-layer-normalization/low-rank-adaptation (AdaLN-LoRA) modulation vector. Learned MLPs M_e and M_r project the descriptor, while $\alpha_s = \lambda m_s$ combines a fixed scale λ with a binary latent-frame mask m_s , as specified in Sec. IV-C:

$$\begin{aligned}e'_{\sigma, s} &= e_{\sigma, s} + \alpha_s M_e(\psi_k), \\ r'_s &= r_s + \alpha_s M_r(\psi_k).\end{aligned}\quad (10)$$

The primed quantities replace their unmodulated counterparts in every block, forming a FiLM-like route through AdaLN [29], [30]. The roughly 100M added parameters amount to 5% of the backbone.

Every context interaction precedes the query. The reference prediction is discarded after subtraction, and the WAM goal marker appears only in the conditioned observation, never in either future being subtracted.

D. DDM-pred: Dense future prediction

Whereas DDM-cond changes how past interactions are represented, DDM-pred determines which dynamics evidence reaches WAM through its training target. World models conventionally predict a temporally dense future, but world-action models often supervise only a terminal outcome. Cosmos Policy, for example, repeats the terminal observation to fill its future slot [2]. Within the same architecture and spatiotemporal encoder, DDM-pred changes only this target, replacing terminal repeats with observations distributed across the prediction horizon:

$$z_{t, H} = \mathcal{E}\left((o_{t+\lfloor qH/4 \rfloor})_{q=1}^4\right). \quad (11)$$

The resulting short trajectory preserves motion that may disappear after the object settles while occupying the existing future component of $\mathbf{x}_0$ in Eq. (2); it therefore changes neither the latent budget nor the action-generation interface.



Fig. 5. **Selected future-prediction examples.** Panels show push-digit and toss-cube results; within each, the realized future is left, Cosmos-Predict2 and DDM predictions are center-top and center-bottom, and their error maps are right. Futures are evaluated at 1.40 s (push) and 1.20 s (toss), with identical crops within each task. Darker maps indicate larger mean absolute RGB error (8-bit units; common scale saturated at 64). In both examples, DDM more closely localizes the realized object and produces a visibly weaker error map than the dynamics-blind reference.

TABLE I

PUSH-DIGIT WM MODE RESULTS.

Method	MSE ↓	Dynamics ID (%) ↑		
		Train (4)	All (5)	Held-out
Cosmos-Predict2	—	25.00	—	—
IIDA+, $k = 2$	0.0184	28.00	21.67	—
IIDA+, $k = 4$	0.0181	29.96	21.87	—
IIDA+, $k = 8$	0.0171	31.96	23.83	16.83
DDM, $k = 2$	0.0158	37.58	29.00	—
DDM, $k = 4$	0.0133	41.46	32.20	—
DDM, $k = 8$	0.0125	44.25	34.90	24.33

TABLE II

TOSS-CUBE WM MODE RESULTS.

Method	WMSE ↓	Yaw ID (%) ↑
Cosmos-Predict2	0.0437	14.29
IIDA+, $k = 2$	0.0268	63.69
IIDA+, $k = 4$	0.0269	60.71
IIDA+, $k = 8$	0.0269	62.50
DDM, $k = 2$	0.0257	63.69
DDM, $k = 4$	0.0255	63.69
DDM, $k = 8$	0.0250	66.67

We use DDM-pred to name this target change in WAM; WM already uses the same dense target by default. Unless stated, DDM uses DDM-cond in both modes, retains standard dense supervision for WM, and applies DDM-pred to WAM. Each WAM reference is a context-free Cosmos Policy instance matched to its arm’s future target: the default dense model and the repeated-terminal ablation therefore use separately trained references and residual stores. Within each store, realized and reference futures use the same encoding.

IV. EXPERIMENTS

Our matched counterfactuals remove visual cues to dynamics. For an identical query, a WM must resolve dynamics-dependent futures, whereas a WAM must generate dynamics-appropriate actions. The two domains use separate banks and metrics, with comparisons made only within each protocol.

A. Tasks and counterfactual datasets

Task 1: planar pushing under latent inertial dynamics.

A cylindrical pusher strikes a seven-segment digit whose invisible ballast places 90% of its mass at one of five positions spaced 1.75 cm apart. Thus, identical state-action queries yield different translations and rotations. Given a common goal, WAM generates a three-parameter primitive whose relative approach angle, speed, and lateral offset control push direction, travel rate, and perpendicular displacement, respectively. Component-wise normalization gives $a_t \in [-1, 1]^3$

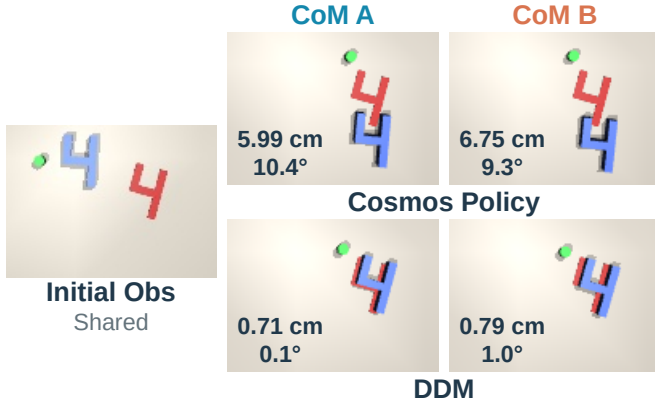
(Fig. 4). Models train on four ballast positions; WM additionally evaluates $y+1.75$ as unseen dynamics, while WAM evaluates held-out goal windows over the training dynamics.

Task 2: dynamic tossing under latent actuation dynamics. A Franka swings and releases an 8 cm cube under 35 visually indistinguishable combinations of swing- and yaw-joint damping. WM predicts the resulting damping-dependent flight; WAM controls a shared landing goal through release time and a yaw command that steers lateral launch direction. Their normalized primitive is $a_t \in [-1, 1]^2$ (Fig. 3). The WM split withholds one swing-damping row and identifies its seven yaw-damping values. WAM evaluates unseen goals over 24 trained regimes and unseen dynamics over 11 held-out regimes.

RoboCasa dense-target check. Outside controlled dynamics, we isolate DDM-pred on 24 RoboCasa tasks, including pick-and-place [31]. The comparison changes only Cosmos Policy’s target to Eq. (11); both arms train for 100,000 iterations on eight H100 GPUs. We report per-task and macro-average success from this descriptive comparison.

Controlled data come from MuJoCo counterfactual banks [32], which replay each serialized transition under every dynamics value. The push WM bank contains 200 shards of 120 transitions, split 160/40 by shard. Each query receives $k \in \{2, 4, 8\}$ distinct, preceding same-regime interactions, grouped only after the split and without identity or dynamics labels. DDM and matched history receive identical context-query pairs.

(C) Push Digit World Action Model Task



(D) Toss Cube World Action Model Task

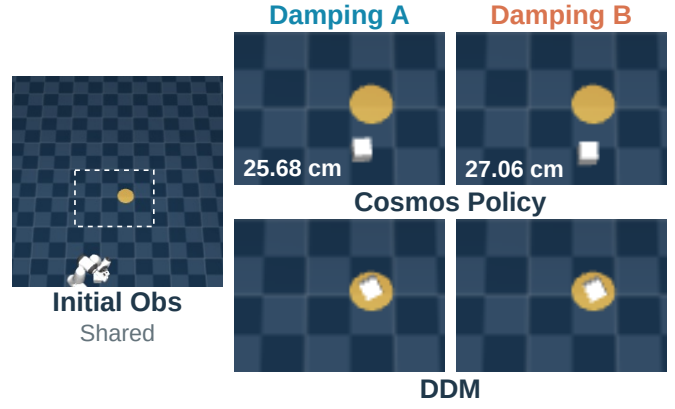


Fig. 6. **Selected goal-reaching examples.** Shared start and goal are shown left; Cosmos Policy and DDM outcomes in the top/bottom rows under dynamics A/B (center/right). Push: terminal observations after one open-loop action, with position/yaw error (cm/degrees); red target, green pusher. Toss: first-contact replays of snapped actions, with actual distance to the yellow goal center (cm), not regret. The dashed box marks the matched outcome crop. Across both dynamics in these examples, DDM reaches substantially closer to the shared goal than the dynamics-blind Cosmos Policy.

TABLE III
PUSH-DIGIT WAM MODE RESULTS.

	Success (%) $\uparrow$	Δ vs CP (pp)
Cosmos Policy (matched)	45.25	—
IPA Policy+, $k = 2$	46.37	+1.1
IPA Policy+, $k = 4$	45.75	+0.5
IPA Policy+, $k = 8$	46.75	+1.5
DDM, $k = 2$	55.13	+9.9
DDM, $k = 4$	55.75	+10.5
DDM, $k = 8$	57.63	+12.4
DDM $k = 8$, RDD= 0	45.25	—
Dynamics oracle	75.12	+29.9

TABLE IV
TOSS-CUBE WAM MODE RESULTS.

Method	Landing regret (cm) $\downarrow$	
	Unseen goals	Unseen dynamics
Cosmos Policy (matched)	10.42	7.08
IPA Policy+, $k = 2$	1.29	3.03
IPA Policy+, $k = 4$	1.29	3.05
IPA Policy+, $k = 8$	1.36	3.05
DDM, $k = 2$	1.40	3.08
DDM, $k = 4$	1.36	3.03
DDM, $k = 8$	1.38	2.95

B. Metrics and baselines

Dynamics identification. Let $\mathcal{V}$ contain the tested regimes, $\hat{z}_v$ the prediction conditioned on v , and z_w^* the counterfactual future under w . For $d_{v,w} = \|\hat{z}_v - z_w^*\|$, we report $\Pr[\arg \min_{w \in \mathcal{V}} d_{v,w} = v]$, with chance $1/|\mathcal{V}|$. Matched variants share batch position and sampler noise, making blind predictions exactly chance. This measures sensitivity rather than named-parameter recovery. Push tests four training or all five regimes; toss tests seven yaw-damping values within the held-out swing-damping row.

Future-prediction error. Push uses per-element latent mean-squared error (MSE). Toss reports windowed latent MSE (WMSE) within a 14×14 region centered identically for all arms on the true cube, preventing the static background from dominating the score.

Push-digit action success. One sampled parameterized action is executed open-loop and succeeds within 1.5 cm and 2° of the goal; expert replay succeeds on 99.75% of rows. Success rates are percentages and differences use percentage points (pp).

Toss-cube landing regret. For each goal and regime, landing regret subtracts the best bank-action landing distance from the policy’s landing-to-goal distance. Generated actions are snapped to the nearest pre-rolled action before scoring. We report mean regret in centimeters; the lookup protocol has an empirical floor of 1.2 cm.

Cosmos-Predict2 and Cosmos Policy (CP). These dynamics-blind WM and WAM references, respectively, receive no interaction context. The matched Cosmos Policy uses the same DDM-pred supervision as DDM but no RDD, isolating residual dynamics conditioning.

IIDA+ and IPA Policy+. These enhanced history baselines adapt IIDA’s transition conditioning [7] to WM and IPA Policy’s interaction conditioning [9] to WAM. Both encode realized futures rather than residuals and match DDM’s k interactions, reference warm start, context encoder, optimization, and evaluation. The suffix denotes our larger pretrained adaptation.

Dynamics oracle. Ground-truth dynamics enter through the same modulation interface, exposing the remaining gap to perfect dynamics context without changing the WAM consumer.

C. Implementation and evaluation protocol

All modes use 224×224 inputs. Reported model and frozen-reference sampling use the training-range EDM schedule ($\sigma_{\max} = 200$, $\sigma_{\min} = 0.01$), guidance 1.5, and 20 denoising steps for push WM or 35 otherwise. A common fixed scale λ is used throughout; push modulates all frames, whereas toss modulates generated frames only. All methods use the same fixed held-out counterfactual queries. Within each task and mode, DDM and matched history share λ , m_s ,

TABLE V
ROBOCASA WAM MODE RESULTS.

Task	CP (%)	DDM-pred (%)	Δ (pp)	Task	CP (%)	DDM-pred (%)	Δ (pp)	Task	CP (%)	DDM-pred (%)	Δ (pp)
PnP Counter→Cab.	22.0	17.3	-4.7	Open single door	42.7	48.0	+5.3	Turn on sink faucet	70.7	46.7	-24.0
PnP Cab.→Counter	9.3	12.7	+3.4	Close single door	64.7	81.3	+16.6	Turn off sink faucet	72.7	79.3	+6.6
PnP Counter→Sink	16.0	23.3	+7.3	Open double door	48.7	71.3	+22.6	Turn sink spout	78.0	78.7	+0.7
PnP Sink→Counter	25.3	27.3	+2.0	Close double door	45.3	52.0	+6.7	Coffee: setup mug	8.7	8.7	0.0
PnP Counter→Micro.	4.7	10.7	+6.0	Open drawer	44.0	51.3	+7.3	Coffee: serve mug	32.7	34.0	+1.3
PnP Micro.→Counter	13.3	19.3	+6.0	Close drawer	93.3	95.3	+2.0	Coffee: press button	40.0	54.7	+14.7
PnP Counter→Stove	6.7	11.3	+4.6	Turn on stove	34.0	36.0	+2.0	Turn on microwave	33.3	32.0	-1.3
PnP Stove→Counter	9.3	12.0	+2.7	Turn off stove	17.3	16.7	-0.6	Turn off microwave	77.3	58.0	-19.3
Mean (24 tasks)	37.9	40.7	+2.8	Improved / tied / lower: 18 / 1 / 5							

descriptor dropout/noise, warm start, schedule, and iteration-matched checkpoints. RDD= 0 uses the same checkpoint, each fixed reference caches one sample per interaction, and external baselines use early stopping.

D. Model-relative residuals identify the active dynamics

On push-digit WM, prediction residuals provide more useful dynamics context than the same transitions encoded directly (Table I). At $k = 8$, DDM raises identification over matched history from 31.96% to 44.25% (+12.29 pp; chance: 25.00%). Across $k \in \{2, 4, 8\}$, DDM lowers per-element latent MSE from 0.0184, 0.0181, 0.0171 to 0.0158, 0.0133, 0.0125, respectively. Figure 5 shows selected large localization improvements for push ballast $y=-1.75$ cm and toss ($b_{\text{swing}}, b_{\text{yaw}} = (13, 10)$). Pixel error is distinct from the latent RDD; examples are illustrative.

E. Residual context extends beyond training dynamics

The RDD is not tied to a catalog of training labels. At $k = 8$, adding the held-out $y+1.75$ regime raises DDM identification to 34.90%, versus 23.83% for history and 20.00% chance; on that slice alone, DDM scores 24.33% versus 16.83%. The value is excluded from training, while the query is conditioned on interactions from the same held-out regime, showing that residual context remains informative during few-shot adaptation beyond the training dynamics.

F. Dense targets preserve transient dynamics evidence

Dense future targets expose dynamics evidence along the trajectory rather than only after the object settles. For toss-cube WM with DDM at $k = 4$, yaw-damping identification is 63.69% from the four-frame future but only 17.26% from its terminal frame (Cosmos-Predict2: 14.29%), localizing the useful evidence to the motion. The direct target comparison is more conservative: at the shared $t = 1.2$ s frame, dense and repeated-terminal training are unresolved in raw pixel error, while correcting each target for its VAE reconstruction floor favors dense prediction on 101/168 rows.

RoboCasa serves only as an external target-design check, without DDM-cond or controlled dynamics shifts. On this fixed 24-task benchmark, dense targets improve 18 tasks, tie one, and increase the reported macro-average from 37.9% to 40.7% (+2.8 pp; Table V). This descriptive result suggests that dense future supervision can also benefit standard manipulation beyond our dynamics-sensitive settings.



Fig. 7. **Real-world chair-pushing check.** Two successful Cosmos Policy rollouts with DDM-pred (top/bottom) progress left to right. Across ten trials per method, DDM-pred succeeds in 7/10 versus 5/10.

Real-world validation. A Fauna Sprout pushes an omnidirectional-wheeled chair from random poses to a wall; success requires reaching it while remaining largely oriented toward it. Cosmos Policy with DDM-pred obtains 7/10 successes versus 5/10 without it (Fig. 7).

G. Model-relative context changes the generated action

At $k \in \{2, 4, 8\}$, DDM improves over IPA Policy+ by 8.76-10.88 pp (Table III). At $k = 8$, DDM reaches 57.63%, or +12.38 pp over matched Cosmos Policy.

With the same WAM interface, injecting the true center-of-mass regime raises success to 75.12%, confirming the value of dynamics context. DDM recovers a substantial share from interactions alone, without dynamics labels; zeroing the RDD of the same $k = 8$ checkpoint returns success to 45.25%, a 12.38 pp reduction.

On the extreme-dynamics slice, DDM exceeds IPA Policy+ by +10.50 pp. A repeated-terminal ablation finds DDM above matched history by +5.00 pp at $k = 2$ and +12.50 pp at $k = 8$. Because dense and repeated-terminal arms use separately trained, target-specific references, their absolute scores are parallel diagnostics rather than a controlled target-only comparison. Figure 6 visualizes representative push and toss outcomes generated from shared starts and goals.

H. Interaction conditioning transfers to actuation dynamics

Toss-cube asks whether the same interface transfers from an object’s latent inertia to the robot’s latent actuation dynamics. In WAM mode (Table IV), interaction conditioning cuts unseen-goal landing regret from 10.42 cm to 1.29-1.40 cm and unseen-dynamics regret from 7.08 cm to 2.95-3.08 cm, winning 386-398 of 480 unseen-goal pairs. The columns are not directly ordered because unseen-goal uses held-out locations, whereas unseen-dynamics retains training goals. For the two evaluated history checkpoints, zeroing removes 71-75% of the gain. Across evaluated k values, both

representations remain near the 1.2 cm candidate-bank floor on unseen goals, differing by at most 0.11 cm. Toss-cube therefore establishes transfer to latent actuation dynamics, while push-digit provides the more discriminative residual-versus-history comparison.

Table II reports the corresponding WM comparison. At $k = 8$, DDM obtains 0.0250 WMSE and 66.67% identification, versus 0.0269 and 62.50% for history.

Within the context action range, the WMSE difference is +0.0036, with DDM lower on 68 of 84 matched rows, indicating more effective use of action-relevant interaction evidence.

V. CONCLUSION

DDM makes video-based prediction and action generation responsive to latent dynamics. DDM-cond distills persistent mismatch from a frozen reference into context, while DDM-pred preserves the transient evidence used to learn it. On push-digit, residual context improves identification over matched history and raises open-loop success over Cosmos Policy; toss-cube transfers the interface from latent inertia to actuation dynamics. The remaining oracle gap and the eight interactions used by our strongest push setting motivate extracting more precise context from fewer interactions.

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